BCA Semester 1 Examination 2025

Model Solutions: 1BCA1 - Fundamentals of Computer

Session	Paper Code	Time	Full Marks	Pass Marks
2024-2027	111601	3 Hours	100	40

Instructions: Answer any five questions (equal marks). Solutions for all questions are given below for practice. Write answers in your own words in the exam.

Q1. What is Disk Operating System? How does it manage files? Discuss with file related commands.

Disk Operating System (DOS)

DOS is a single-user, single-tasking, command-line (character interface) operating system that is loaded from a disk. MS-DOS (Microsoft Disk Operating System) was the most popular version for IBM PCs. The user types commands at the prompt (C:\>) and the command interpreter (COMMAND.COM) executes them. Main system files: IO.SYS, MSDOS.SYS and COMMAND.COM.

File management in DOS

- **File naming:** every file has a name (up to 8 characters) and an extension (up to 3 characters), e.g. REPORT.TXT. Extensions show the type: .EXE/.COM/.BAT (programs), .TXT, .DOC, .BAS.
- **Directory structure:** files are grouped into directories (folders) arranged as a tree starting at the root directory (\).
- **File Allocation Table (FAT):** the disk is divided into tracks, sectors and clusters; the FAT records which clusters belong to each file and which are free, so DOS can find, extend and delete files.
- **Wildcards:** * (any group of characters) and ? (any single character), e.g. *.TXT, A?.DOC.
- **Attributes:** read-only, hidden, system, archive; they protect files from accidental change.

File related commands

Command	Type	Purpose and example
DIR	Internal	Lists files and directories. DIR *.TXT /P
COPY	Internal	Copies files. COPY A.TXT D:\BACKUP; COPY CON FILE.TXT creates a file from the keyboard
REN / RENAME	Internal	Renames a file. REN OLD.TXT NEW.TXT
DEL / ERASE	Internal	Deletes files. DEL *.BAK
TYPE	Internal	Displays the contents of a text file. TYPE NOTES.TXT
MD / CD / RD	Internal	Make, change and remove directories. MD STUDENT
XCOPY	External	Copies files with sub-directories. XCOPY C:\DATA D:\DATA /S
MOVE	External	Moves files. MOVE A.TXT D:\
ATTRIB	External	Shows or sets attributes. ATTRIB +R FILE.TXT
EDIT	External	Opens the text editor to create/modify files. EDIT FILE.TXT
FIND / COMP	External	Searches text in a file / compares two files

Q2. Differentiate between any two of the following

(i) Single user vs Multi-user system

Single-user system	Multi-user system
Only one user can use the computer at a time.	Many users can use the system at the same time through terminals.
Resources are used by one person.	Resources (CPU, memory, printers) are shared among users.
Simple, low cost, little security needed.	Complex, costly, needs user accounts and strong security.
Examples: MS-DOS, Windows 98.	Examples: UNIX, Linux server, mainframe OS.

(ii) Decimal vs Binary number system

Decimal	Binary
Base 10; digits 0 to 9.	Base 2; digits 0 and 1.
Place values are powers of 10.	Place values are powers of 2.
Used by humans in daily life.	Used internally by computers (ON/OFF states).
Example: (25)10	Example: (11001)2

(iii) 9's vs 10's complement

9's complement	10's complement
Obtained by subtracting each digit from 9.	Obtained by adding 1 to the 9's complement.
Example: 9's complement of 32 = 67.	Example: 10's complement of 32 = 68.
End-around carry is added to the result.	Final carry is simply discarded.
Has two representations of zero (00 and 99).	Has one representation of zero.

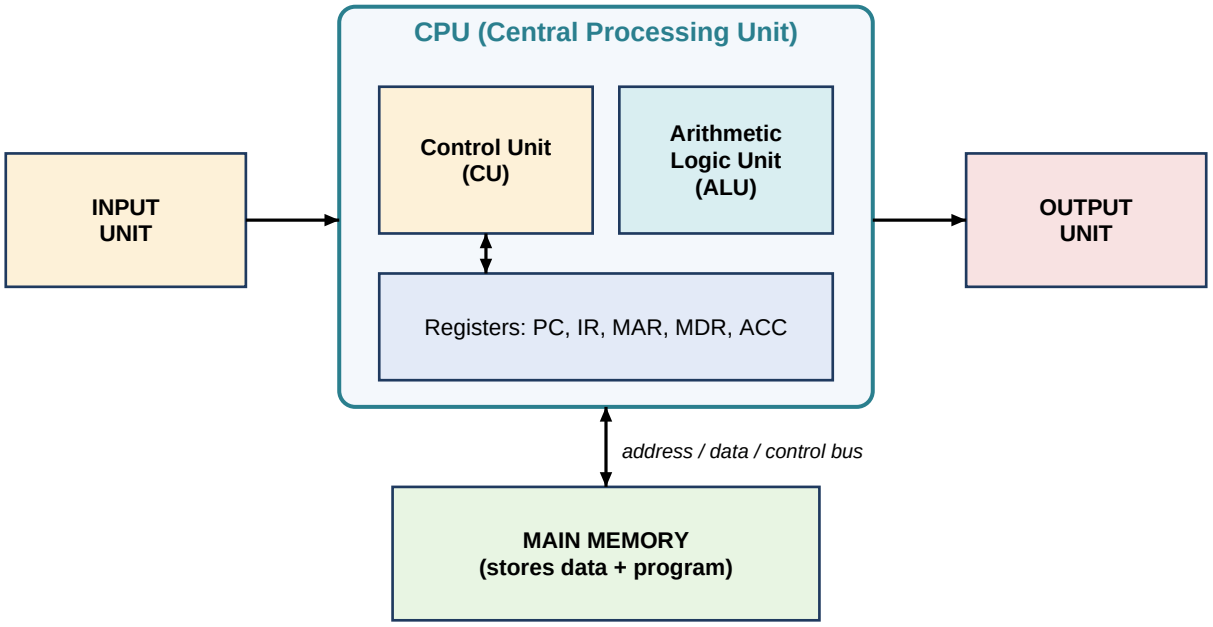
(iv) Static RAM vs Dynamic RAM

Static RAM (SRAM)	Dynamic RAM (DRAM)
Uses flip-flops; needs no refresh.	Uses capacitor and transistor; needs periodic refresh.
Faster and more expensive.	Slower and cheaper.
Low density, high power use.	High density, lower power use.
Used as cache memory.	Used as main memory.

Q3. With pictorial representation discuss Von Neumann architecture of computer and its working behaviour.

In 1945 John von Neumann proposed the **stored-program concept**: both the program instructions and the data are stored together in the same main memory, and the CPU fetches and executes instructions one after another. Almost all modern computers follow this model.

Block diagram



Components

- **Input unit:** sends data and instructions into the computer (keyboard, mouse).
- **Memory unit:** stores the program and data; each location has an address.
- **Control Unit (CU):** fetches, decodes and controls execution of instructions and directs all other units.
- **ALU:** performs arithmetic (+ - * /) and logical (AND, OR, NOT, comparison) operations.
- **Registers:** PC (program counter - address of next instruction), IR (instruction register), MAR (memory address register), MDR (memory data register), ACC (accumulator).
- **Output unit:** shows results (monitor, printer). **Buses:** address, data and control lines connect the units.

Working (instruction cycle)

1. **Fetch:** the address in PC is placed in MAR; the instruction is read from memory into MDR and then IR; PC is incremented.
2. **Decode:** the control unit decodes the instruction in IR to find the operation and operands.
3. **Execute:** the ALU performs the operation, reading operands from memory/registers if required.
4. **Store:** the result is written to a register or memory, or sent to the output unit; the cycle repeats until HALT.

Limitation (Von Neumann bottleneck): because instructions and data share one bus, the CPU cannot fetch both at the same time, which limits speed.

Q4. Discuss Digital and Analog computer with their features and examples. Also write some major differences.

Digital computer

A digital computer processes discrete data represented in binary form (0 and 1) and counts the digits to calculate.

- Features: high accuracy, high speed, large storage, general-purpose, programmable, works on discrete signals.
- Examples: desktop PC, laptop, mainframe, smartphone, digital watch, calculator.

Analog computer

An analog computer processes continuous physical quantities such as voltage, temperature, pressure and speed, and gives the result by measurement of these quantities.

- Features: less accurate (limited by instrument reading), very fast for real-time simulation, little storage, usually special purpose.
- Examples: speedometer, mercury thermometer, petrol pump flow meter, seismograph, ECG machine.

Hybrid computer

Combines both: analog input is converted to digital for processing, e.g. ICU patient monitors and weather prediction systems.

Basis	Digital computer	Analog computer
Data	Discrete (binary digits)	Continuous physical quantities
Accuracy	High	Low
Storage	Large	Little or none
Purpose	General purpose	Special purpose
Operation	Counting of digits	Measuring
Cost	Cheaper per function today	Costlier for the same job
Examples	PC, laptop	Speedometer, thermometer

Q5. Explain different binary arithmetic operations using 1's and 2's complements with example.

Computers have no separate subtraction circuit; subtraction A - B is done by **adding the complement of B** to A.

Finding the complements

- **1's complement:** change every 0 to 1 and every 1 to 0. Example: 0011 becomes 1100.
- **2's complement:** 1's complement + 1. Example: 0011 becomes 1100 + 1 = 1101.

A. Subtraction using 1's complement (rule: add; carry out means positive)

Example 1 (positive result): 7 - 3 (4-bit). 7 = 0111, 3 = 0011; 1's complement of 3 = 1100.

```
  0111
+ 1100
-------
1 0011   -> carry generated, so add the carry back (end-around carry)
  0011 + 1 = 0100 = +4
```

Example 2 (negative result): 3 - 7. 1's complement of 7 = 1000.

```
  0011
+ 1000
-------
  1011   -> no carry, so the result is negative
  answer = -(1's complement of 1011) = -(0100) = -4
```

B. Subtraction using 2's complement (rule: carry out is discarded)

Example 1: 7 - 3. 2's complement of 3 = 1101.

```
  0111
+ 1101
-------
1 0100   -> discard the carry, result = 0100 = +4
```

Example 2: 3 - 7. 2's complement of 7 = 1001.

```
  0011
+ 1001
-------
  1100   -> no carry, so the result is negative
  answer = -(2's complement of 1100) = -(0100) = -4
```

C. Binary addition and multiplication/division

- **Addition rules:** 0+0=0, 0+1=1, 1+0=1, 1+1=10 (0 carry 1), 1+1+1=11. Example: 1011 + 0110 = 10001 (11 + 6 = 17).
- **Multiplication:** shift and add. 101 x 11 = 101 + 1010 = 1111 (5 x 3 = 15).
- **Division:** repeated subtraction/shift. 1100 / 11 = 100 (12 / 3 = 4).

Feature	1's complement	2's complement
Carry out	Added back (end-around carry)	Discarded
Zero	Two forms (+0, -0)	One form
Range (n bits)	-(2^(n-1) - 1) to +(2^(n-1) - 1)	-2^(n-1) to +(2^(n-1) - 1)
Use today	Rare	Standard in computers

Q6. Convert the following numbers to decimal

(i) $(ACD)_{16}$

A = 10, C = 12, D = 13. = $10x16^2 + 12x16^1 + 13x16^0$ = 2560 + 192 + 13 = **2765**

(ii) $(1111.11)_2$

Integer part: $1x2^3 + 1x2^2 + 1x2^1 + 1x2^0$ = 8 + 4 + 2 + 1 = 15. Fraction part: $1x2^{-1} + 1x2^{-2}$ = 0.5 + 0.25 = 0.75. Result = **15.75**

(iii) $(756)_8$

= $7x8^2 + 5x8^1 + 6x8^0$ = 448 + 40 + 6 = **494**

(iv) $(CD)_{16}$

= $12x16^1 + 13x16^0$ = 192 + 13 = **205**

(v) $(BCA)_{16}$

= $11x16^2 + 12x16^1 + 10x16^0$ = 2816 + 192 + 10 = **3018**

Number	Decimal
(ACD)16	2765
(1111.11)2	15.75
(756)8	494
(CD)16	205

Number	Decimal
(BCA)16	3018

Q7. Discuss the architecture of Hard Disk Drive. How does it work? Also discuss seek time, latency time and access time.

Architecture

A hard disk drive (HDD) is a non-volatile magnetic secondary storage device. Main parts:

- **Platters:** rigid aluminium/glass disks coated with magnetic material, stacked on a spindle; both surfaces store data.
- **Spindle and motor:** spin the platters at 5400, 7200 or more rpm.
- **Read/write heads:** one per surface, floating a few nanometres above the platter on an air cushion.
- **Actuator arm and voice-coil motor:** move all heads together across the platter.
- **Tracks, sectors, cylinders:** each surface has concentric *tracks* divided into *sectors* (usually 512 bytes or 4 KB); the same track on all platters forms a *cylinder*.
- **Controller board and cache buffer:** interface (SATA) and electronics that manage data transfer.

Working

To access data the controller (1) moves the arm so the head is over the required track (seek), (2) waits for the platter to rotate until the required sector is under the head (rotational latency), and (3) transfers the data: writing magnetises tiny regions of the surface in one of two directions (1 or 0), and reading detects these magnetic changes.

Access-time components

Term	Meaning	Typical value
Seek time	Time taken by the head to move to the required track.	3 to 12 ms (average about 9 ms)
Latency time (rotational delay)	Time for the required sector to rotate under the head. Average = half a rotation = 0.5 x 60 / rpm seconds.	About 4.17 ms at 7200 rpm
Transfer time	Time to read/write the data once positioned.	Very small
Access time	Total time to get data = seek time + latency time (+ transfer time).	About 13 ms

Example: seek time 9 ms and latency 4.17 ms give an access time of about 13.17 ms.